Same bytes, same meaning: how WDF makes documents verifiable for humans and machines

Every WDF package carries two views of one document — a rendered page for people and a Markdown layer for machines — and a proof that they cannot drift apart.

The problem: documents that lie to machines

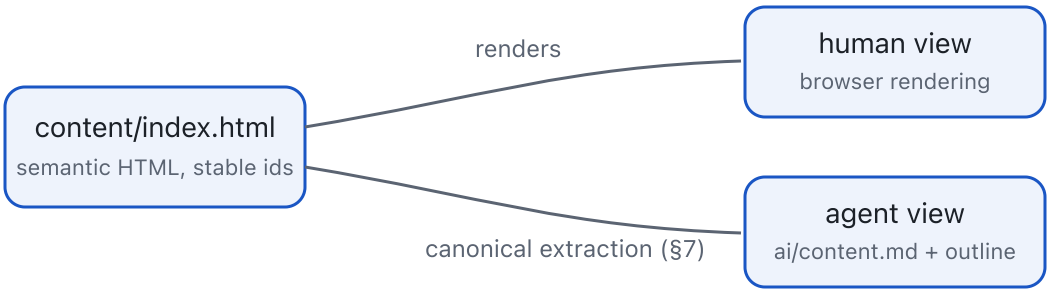
An entire industry exists to reconstruct structure from PDFs before feeding them to language models. The results are probabilistic: headings guessed from font sizes, tables re-assembled from x/y coordinates, citations pointing at page numbers rather than statements. As *IDRsolutions*, a PDF toolmaker, puts it in a member news item on pdfa.org (October 2025), PDFs are “visually focused, not AI-friendly”.

The failure is structural. When the machine-readable view is *derived by heuristics*, nobody can say whether it matches what the human saw — and when money or law is involved, “probably the same” is not an answer.

The WDF approach

WDF inverts the relationship. The package contains constrained semantic HTML as the single source of truth, and a **canonical extraction algorithm** — normative, step by step — derives the machine view from it. Determinism is the contract:

> Same input bytes, same output bytes, on every platform. A validator re-runs the extraction and rejects the package on any difference.



One source, two synchronized views. Tampering with either side is detectable.

Every citable element carries a stable identifier, so an agent can answer with a citation such as `wdf:urn:uuid:…#tbl-spesa-2025` that anyone can resolve and verify — down to the paragraph, H_2O-molecule precision instead of page^2-level guesswork.

What verification looks like

The reference implementation exposes the whole check as one call:

```
import { readPackage, verifyPackage } from '@wdf-dev/core';

const pkg = readPackage(bytes);
const result = await verifyPackage(pkg);
// result.integrity   — every file matches its SHA-256 digest
// result.determinism — ai/ is the canonical extraction of content/
// result.verified    — both, or the package is rejected
```

Note what this rules out: an attacker who edits the Markdown and diligently recomputes every hash still fails, because the *determinism* check re-derives the agent view from the human view and compares byte by byte.

Small glossary

canonical extraction
The normative algorithm (spec §7) mapping WDF-HTML to Markdown and an outline, with no locale-, time- or platform-dependent behavior.

citable element
A section, heading, paragraph, table, figure, blockquote or top-level list item, carrying a mandatory stable `id`.

standalone profile
A single HTML file embedding viewer and package: double-click to open, offline, and the canonical bytes remain extractable.

The specification and tooling are open: repository and WDF Core 0.1 draft. Feedback is welcome as an issue on the repository.